SSP

Computer & Communication Department

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Modern Physics
Sheet 8
Solutions

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1. Calculate the de Broglie wavelength for a proton moving with a speed of 10^6 m/s.

Solution

$v = 10^6$ m/s << c (classical relations can be used)

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{1.67 \times 10^{-27} \times 10^6} = 3.97 \times 10^{-13} m$$

2. Through what potential difference would an electron have to be accelerated to give it a de Broglie wavelength of 1.00×10^{-10} m?

Solution

$$p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{1.00 \times 10^{-10}} = 6.63 \times 10^{-24} kg.m/s = 0.0124\ MeV/c$$

pc = 0.0124 MeV << 0.511 MeV

(classical relations can be used)

$$K = \frac{1}{2}mv^2 = \frac{p^2}{2m} = \frac{\left(6.63 \times 10^{-24}\right)^2}{2 \times 9.11 \times 10^{-31}} = 2.41 \times 10^{-17} J = 151\ eV$$

V = 151 V

3. An electron and a photon each have kinetic energy equal to 50 keV. What are their de Broglie wavelengths?

Solution

Electron

K = 0.05 MeV << 0.511 MeV

(Classical relations can be used)

$$p = \sqrt{2mK} = \sqrt{2 \times 9.11 \times 10^{-31} \times 0.05 \times 10^{6} \times 1.6 \times 10^{-19}}$$
$$= 1.21 \times 10^{-22}\ kg.m/s$$

$$\lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34}}{1.21 \times 10^{-22}} = 5.49 \times 10^{-12}\ m$$

Photon

E = 0.05 MeV

p = E/c = 0.05 MeV/c = 2.67×10^{-23} kg.m/s

$$\lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34}}{2.67 \times 10^{-23}} = 2.49 \times 10^{-11}\ m$$

4. To "observe" small objects, one measures the diffraction of particles whose de Broglie wavelength is approximately equal to the object's size. Find the kinetic energy (in electron volts) required for electrons to resolve a nucleus of size 10 fm.

Solution

$\lambda = 10^{-14}$ m

$$p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{10^{-14}} = 6.63 \times 10^{-20} kg.m/s = 124.3\ MeV/c$$

pc = 124.3 MeV >> 0.511 MeV

(Relativistic relations must be used)

$$E^2 = (pc)^2 + (mc^2)^2$$

$$E = \sqrt{124.3^2 - 0.511^2} = 124.3\ MeV$$

$$K = E - mc^2 = 123.8\ MeV$$

5. Find the de Broglie wavelength of a ball of mass 0.20 kg just before it strikes the Earth after being dropped from a building 50 m tall.

Solution

For an object in free fall

$$v = \sqrt{2gh} = \sqrt{2 \times 9.81 \times 50} = 31.3\ m/s$$

$$p = mv = 0.20 \times 31.3 = 6.26\ kg.m/s$$

$$\lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34}}{6.26} = 1.06 \times 10^{-34}\ kg.m/s$$

6. A proton has a kinetic energy of 1.0 MeV. If its momentum is measured with an uncertainty of 5.0%, what is the minimum uncertainty in its position?

Solution

K = 1 MeV << 938 MeV

(Classical relations can be used)

$$p = \sqrt{2mK} = \sqrt{2 \times 1.67 \times 10^{-27} \times 1 \times 10^{6} \times 1.6 \times 10^{-19}} = 2.31 \times 10^{-20}\ kg.m/s$$

$$\Delta p = 0.05 \times 2.31 \times 10^{-20} = 1.16 \times 10^{-21}\ kg.m/s$$

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

$$\Delta x_{min} = \frac{\hbar}{2\Delta p} = \frac{6.63 \times 10^{-34}}{2\pi \times 2 \times 1.16 \times 10^{-21}} = 4.55 \times 10^{-14}\ m$$

7. Estimate the kinetic energy of an electron confined within a nucleus of size 1.0×10^{-14} m by using the uncertainty principle.

Solution

$$\Delta x = \frac{1}{2} \times 1.0 \times 10^{-14} = 5 \times 10^{-15}\ m$$

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

$$\Delta p \geq \frac{\hbar}{2\Delta x} \geq \frac{6.63 \times 10^{-34}}{2\pi \times 2 \times 5 \times 10^{-15}} \geq 1.06 \times 10^{-20}\ kg.m/s \geq 20\ MeV/c$$

The uncertainty in the momentum must be smaller than the momentum itself so:

$p \geq 20$ MeV/c

$$E_{min} = \sqrt{(p_{min}c)^2 - (mc^2)^2} = \sqrt{20^2 - 0.511^2} = 20\ MeV$$

$$K_{min} = E_{min} - mc^2 = 20 - 0.511 = 19.5\ MeV$$

The energy of electrons emitted in beta decay is much smaller than this value, so this proves that no electrons live inside the nucleus.

8. A charged pi meson has a rest energy of 140 MeV and a lifetime of 26 ns. Find the minimum energy uncertainty of the pi meson.

Solution

$$\Delta E \Delta t \geq \frac{\hbar}{2}$$

$$\Delta E_{min} = \frac{\hbar}{2\Delta t} = \frac{6.63 \times 10^{-34}}{2\pi \times 2 \times 26 \times 10^{-9}} = 2.03 \times 10^{-27} J = 1.27 \times 10^{-8}\ eV$$